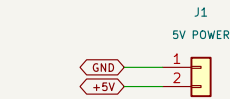
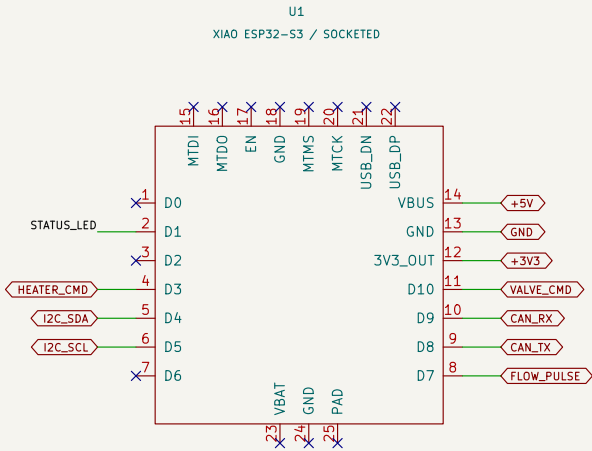


1. POWER – external regulated 5V / XIAO onboard 3.3V regulator

2. XIAO ESP32–S3 – two female 1x7 sockets, 2.54mm pitch



J1 feeds XIAO 5V and the existing 5V interfaces.
XIAO 3V3 supplies CAN VIO, I2C, TMP102 and FLOW pull–up.
USB–C, LDO, BOOT and RESET are on the removable XIAO.
Disconnect external J1 power before USB flashing.

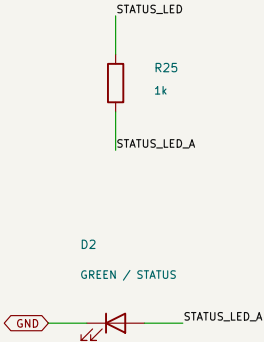


U1: D3/GPIO4 BOILER; D1/GPIO2 LED; D4/GPIO5 SDA; D5/GPIO6 SCL.
D7/GPIO44 FLOW; D8/GPIO7 CAN TX; D9/GPIO8 CAN RX; D10/GPIO9 VALVE.

Seed numbering: 1..7 = D0..D6; 8..11 = D7..D10.
12=3V3 OUT; 13=GND; 14=5V IN. Pins 15..25 are bottom pads, NC.
D0/GPIO1, D2/GPIO3 and D6/GPIO43 are unused.

Footprint: Seeed_XIAO:XIAO–ESP32–S3–DIP.
Fit two female 1x7 sockets; XIAO fitted manually from stock.
Row spacing 15.24mm; 2.54mm pitch. GPIO4 avoids boot straps.

3. STATUS LED – GPIO2 / D1, HIGH = ON



Existing green LED and 1k resistor retained.

FUTURE PCB FIRMWARE: BOILER GPIO4, active HIGH (MOSFET driver).
Preload LOW before output enable; disable heater pull–up.
Current HW–399 firmware uses GPIO3, active LOW; do not use unchanged.
Other existing XIAO pins retained; add status LED on GPIO2.
Firmware unchanged; PCB placement and routing pending.

Sensors interfaces



File: sensors_interfaces.kicad_sch

Socketed XIAO ESP32–S3 / Seeed DIP footprint

CoffeeFlow

Sheet: /

File: sensors.kicad_sch

Title: Sensors – XIAO ESP32–S3 sockets and status LED

Size: A3

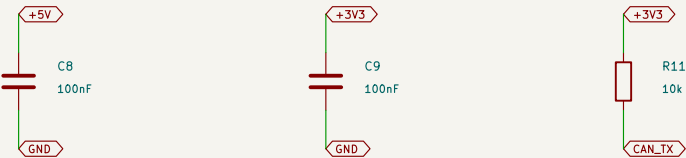
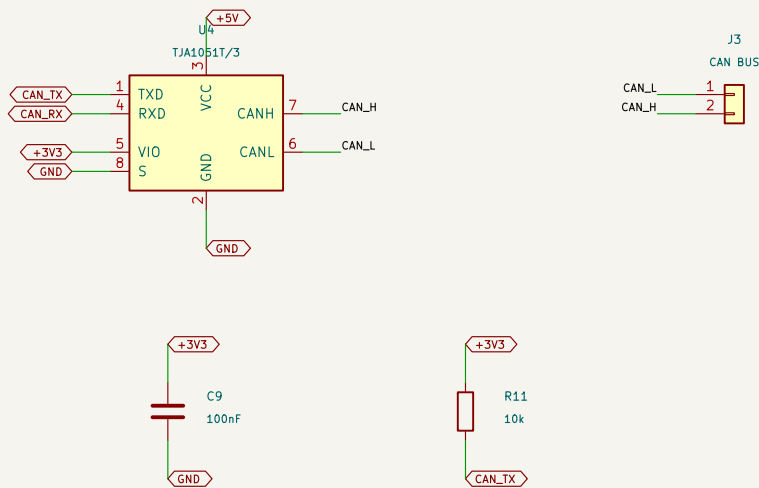
Date: 2026–10–07

Rev: B

KiCad E.D.A. 10.0.6

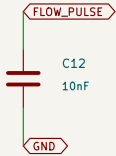
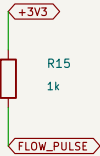
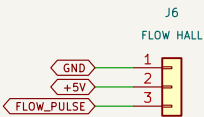
Id: 1/2

4. CAN – TJA1051T/3, 5V bus driver / 3.3V logic



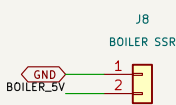
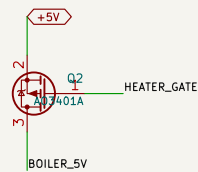
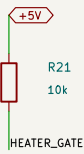
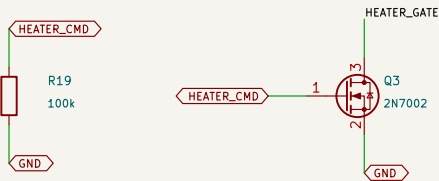
S/SLNT hard-wired low. R11 keeps TX recessive during reset. R12 permanently terminates this bus-end node (120 ohms). J3 has no GND pin: power GND must run alongside CAN pair.

6. DIGMESA – open collector, existing RC filter

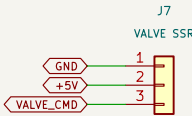
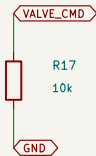


GPIO44 / D7 input; internal pull-up disabled. 1k x 10nF = 10us. Digmesa powered from 5V; signal pulled to 3.3V only. Pin order: GND brown / 5V green / SIGNAL yellow. RC discharge speed also depends on sensor/cable impedance.

8. BOILER SSR – powered 5V output, HIGH command = ON

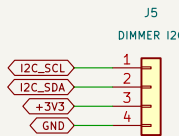
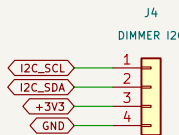
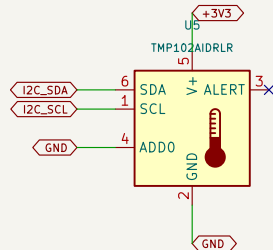
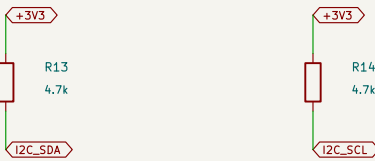


7. VALVE SSR – 3.3V signal / 5V supply



GPIO9 / D10, 3.3V signal / 5V VCC; default LOW.

5. SHARED I2C – 3.3V, Grove/HY2.0 cable order



Grove opening up: left-to-right GND / 3V3 / SDA / SCL. Numbered pads: 1=SCL yellow, 2=SDA white, 3=3V3 red, 4=GND black. Addresses: dimmer 0x50 / XDB401 0x7F / TMP102 0x48. TMP102 ADD0 grounded; ALERT unused. Mount away from XIAO and MOSFET heat. Pull-ups always fitted: 2.35k effective with XDB401 attached. 100kHz initial bus speed; remote modules provide local supply decoupling.

XIAO sockets – schematic only; PCB pending

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Sheet: /Sensors interfaces/

File: sensors_interfaces.kicad_sch

Title: Sensors – CAN, I2C, flowmeter and SSR outputs

Size: A3

Date: 2026-10-01

Rev: A

KiCad E.D.A. 10.0.6

Id: 2/2

J8: SSR (-) -> pin1 GND; SSR (+) -> pin2 switched 5V. Q3 pulls Q2 gate LOW; Q2 supplies the SSR current. GPIO4 / D3 -> HEATER_CMD (HIGH = ON). R19/R21/R22 ensure off with GPIO floating or 3V3 absent; Shared GND. Gate pull-up current about 0.5mA when ON; load <=100mA; no mains.